

GloU Visualisation *Tool*

An interactive 3D application for creating, manipulating and comparing cuboids while computing and visualising GloU in real time.

One product story, three demo handoffs.

NATE · INTRO + DEMO 1

Project definition

Project team, topic, main goals, target users, then object creation and manipulation.

TIM · USER NEEDS + DEMO 2

Use-case proof

Use cases, background research, use-case testing approach, 2D/3D modes, Analyse workflow and metric explanation.

RYAN · DEMO 3 + CONCLUSION

Status and lessons

Save/load, extra features, achieved and remaining scope, issues encountered, teamwork and project management learning.

TEAM

TOPIC

USERS

TESTING

DEMO

CONCLUSION

What are IoU and GloU?

IOU INTERSECTION OVER UNION

How much do two boxes overlap?

IoU compares the shared area or volume against the total combined area or volume of two objects.

- High IoU means strong overlap.
- Zero IoU means no overlap.
- Useful, but silent when boxes are separate.

GIOU GENERALISED IOU

How good is the spatial match?

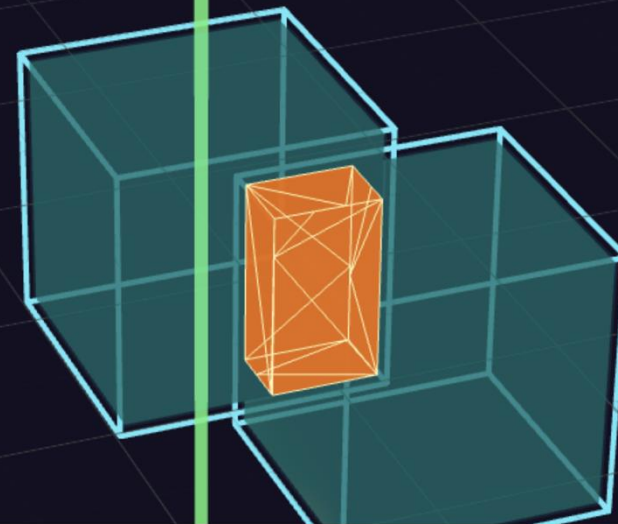
GIoU extends IoU by adding a penalty based on the smallest enclosing box, so it remains meaningful even when objects do not overlap.

- Considers overlap and enclosure.
- Changes even in non-overlap cases.
- Better for teaching spatial metric behaviour.

In this presentation, the key learning loop is: move or resize cuboids → observe IoU/GIoU change → inspect the explanation in Analyse.

PROJECT TOPIC

Interactive IoU / GloU visualisation
for 2D and 3D object comparison.



The main goal is to make object-overlap metrics teachable, visible and repeatable through direct manipulation, live calculation, explanation and scene management.

01

Teach

Help learners understand metric behaviour spatially.

02

Interact

Create, move, rotate and scale objects.

03

Compute

Calculate IoU/GIoU as the scene changes.

04

Explain

Connect values to formulas and selected objects.

05

Repeat

Save and reload scenes for reusable examples.

TARGET USERS

Students.
Educators.
Practitioners

The audience is anyone who needs spatial intuition for object-overlap metrics.

STUDENTS

Computer vision and graphics students who need an intuitive way to explore bounding-box metrics.

EDUCATORS

Lecturers and tutors who need a live teaching aid for overlap, non-overlap and enclosing-box concepts.

PRACTITIONERS

Early-stage CV practitioners exploring object detection metrics and bounding-box regression behaviour.

SOFTWARE DEMO · NATE → TIM → RYAN

A normal user journey, mapped to signed-off RTM evidence.

NATE

Object creation

Add, randomise, select, move, rotate and scale.

TIM

Mode + Analyse

2D/3D views, selected pair, formula and live metric.

RYAN

Persistence

Save/load locally and show extra workflow support.

CLOSE

Status

Achieved, remaining, issues and lessons.

5 SIGNED-OFF RTM ITEMS DEMONSTRATED

Latest RTM IDs are used here.

R2 / R3

Create and transform cuboids.

R1

Calculate IoU/GIoU.

R4 / R5

Display and explain results.

R6

Auto-update values live.

R9

Save and load scenes.

The tool is built around teachable exploration.

01 CLASSROOM DEMONSTRATION

Educator-led explanation

A lecturer projects the tool during an object-detection lesson, creates cuboids and drags them into overlap while students watch the GloU update.

- Shows overlap and non-overlap visually.
- Connects movement to metric change.
- Supports live explanation instead of static diagrams.

02 SELF-PACED EXPLORATION

Student-driven experiments

A student generates cuboid configurations, transforms objects and observes how position, scale and rotation affect IoU/GIoU values.

- Encourages trial-and-error learning.
- Supports repeatable scene setups.
- Can extend to practitioner metric exploration.

The requirement came from a learning gap,
then was tested as a learning loop.

GIOW THEORY

Rezatofighi et al. define GloU for overlap and non-overlap cases using the smallest enclosing box.

IDENTIFIED GAP

Static equations and diagrams do not give beginners enough spatial intuition.

USER REQUIREMENT

Users need manipulation, live values and explanation in one interface.

RTM TRACEABILITY

Each requirement maps to tests that verify functional delivery.

SPRINT REVIEW TESTING

Delivered features are tested and issues are logged before the next cycle.

USER DEMONSTRATION

Target users perform key tasks such as creating cuboids and observing metric changes.

KNOWN-VALUE CHECKS

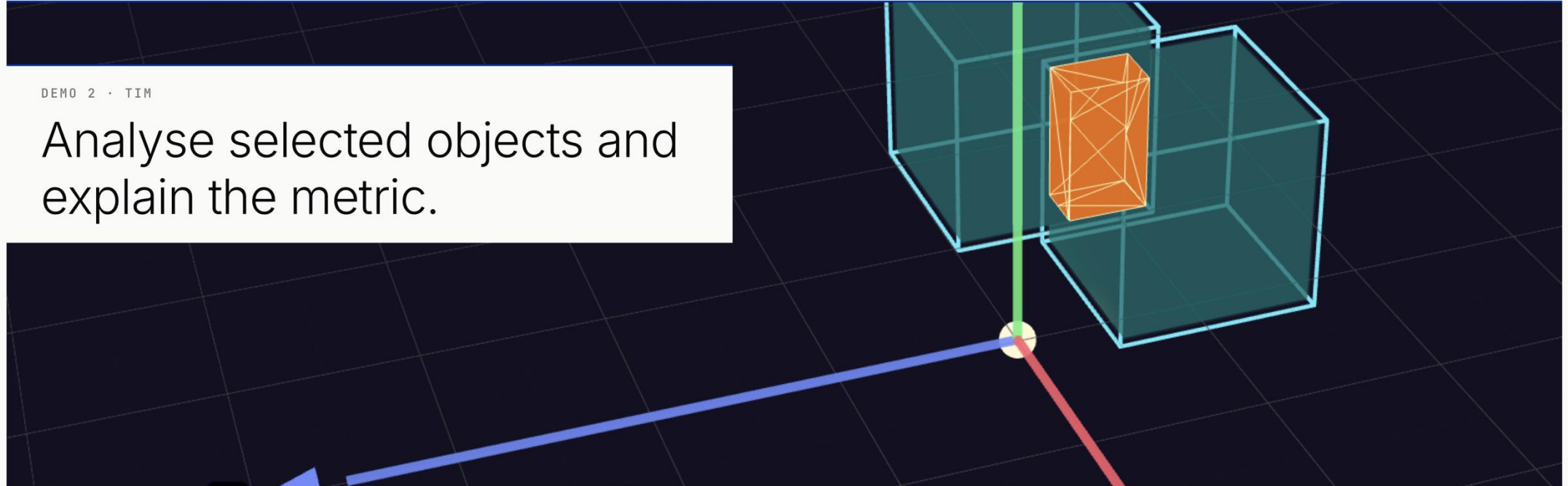
Pre-computed IoU/GIoU examples are compared with tool output.

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The testing approach checks usability, feature delivery and calculation correctness: create object → manipulate object → observe metric change → inspect explanation.

DEMO 2 · TIM

Analyse selected objects and explain the metric.



Switch between 2D and 3D views, then open Analyse to show pair selection, formula explanation, numeric result and live update.

MODE

R11

Switch between 2D and 3D viewing modes.

ANALYSE

R1

Calculate IoU/GIoU for compared objects.

EXPLAIN

R4-R6

Display results and auto-update values live.

A scene can be preserved, restored and continued.

R9 · SAVE LOCALLY

Scene persistence

Save the current cuboid configuration on the computer for repeatable examples.

N6 · LOAD + VALIDATE

Safe recovery

Load the saved scene, confirm objects are restored, and reject malformed scene data.

R10 / N3-N4 . EXTRA

Workflow support

Show import support, clearer controls, status messages and usability improvements where relevant.

RYAN: ACHIEVED / REMAINING / ISSUES

Core goals achieved. *R8/N7 remain.*

The main cuboid, metric, UI, 2D/3D, import and persistence workflows are delivered; only the authentication-related R8 and N7 items are not fully closed.

HONEST SCOPE CLOSURE

STATUS SUMMARY

ACHIEVED / REMAINING / OPEN ISSUES

ACHIEVED / SIGNED-OFF

R1-R7, R9 and R11-R17 are delivered across metric calculation, object interaction, 2D/3D views, visual feedback and save/load.

RECENTLY COMPLETED

R10 and N1-N6 are treated as recently completed polish and support work: import, performance, usability and data reliability.

REMAINING SCOPE / ISSUES

R8 and N7 are authentication-related and not fully closed; open issues are mostly performance, accessibility polish and maintainability cleanup.

CONCLUSION MESSAGE: CORE DELIVERY ACHIEVED, REMAINING WORK BOUNDED.

← → SLIDES · B STATIC · ESC INDEX

Clear ownership mattered most when the team changed.

2

ACTIVE MEMBERS LAST YEAR

The project previously dropped to only two active members, forcing stronger shared awareness.

1

COMMUNICATION LOST THIS YEAR

The remaining team had to stay informed and support each other to avoid delays.

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ISSUES FIXED THIS REVIEW CYCLE

Review findings were converted into implementation items across UI, accessibility, state and interactions.

KEY MESSAGE

Effective teamwork depended on understanding each other's responsibilities and being prepared to support one another when needed.

EXAMPLES FIXED

MetricsTab display behaviour, Analyse mode selection logic, Add Cuboid validation, keyboard shortcuts, accessibility roles and labels, responsive bars, clearer comparison status messages.

RTM + GANTT + RISK REGISTER + REVIEW NOTES

Concrete artefacts made Agile usable.

KEY MESSAGE
Effective project management meant using the RTM, Gantt chart, risk register and review notes as one control system.

ISSUES ENCOUNTERED
Remaining work is bounded: R8/N7 plus performance, accessibility polish and maintainability cleanup.

SIMULATED GANTT VIEW

Built from current RTM status and known sprint progress.

W1-W12



6100 VISUALISATION TOOL . CLOSING

15 / 15

THANK YOU

Questions *welcome.*

We are happy to discuss the demo workflow, RTM coverage, remaining R8/N7 scope, and project management evidence.

NATE TTH RYAN

FINAL PRESENTATION . Q&A
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